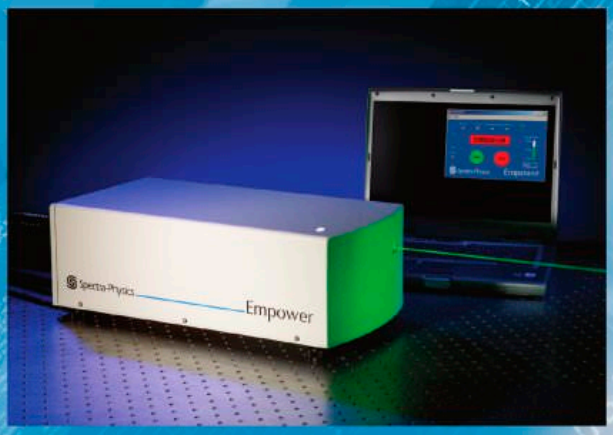
Empower® Q-Switched Laser



The Empower® series is the next generation of high power pulsed green solid state lasers from Spectra-Physics. These Q-switched Nd:YLF lasers deliver an average output power of up to 30 W at 527 nm and pulse repetition rates from 1 kHz to 10 kHz. Ideal for pumping Ti:Sapphire amplifiers such as the Spectra-Physics Spitfire® series, the Empower pulse energies are as high as 20 mJ.

Utilizing diode pumping, the Empower series features excellent output stability with pulse energy variations less than 1% rms. This is a critical advantage resulting in the best energy stability from a Ti:Sapphire ultrafast amplifier. In addition, the Empower output beam guarantees optimum mode-matching with a Ti:Sapphire amplifier, resulting in high conversion efficiencies.

The Empower series' highly integrated design benefits from Spectra-Physics' unmatched diode-pumped solid state laser experience and field-proven components. The ruggedized architecture results in superior overall reliability and lowest cost of ownership.

Although the Empower has been specifically optimized for pumping ultrafast amplifiers, its high pulse energy makes it suitable for other applications, particularly in materials processing.

Empower lasers operate from either 110 V or 230 VAC single-phase power and require no external cooling water. The compact laser head, with the smallest footprint available, together with the fully computer-controlled rack-mountable power supply, simplifies system integration and eliminates the need for special utilities.

The Empower Advantage

- Ideal pump source for Ti:Sapphire amplifiers
- Rugged design for maximum reliability and longevity
- Diode pumping ensures excellent pulse energy stability—better than 1% rms
- 15 W at 5 kHz (Empower 15) or 30 W at 5 kHz (Empower 30)
- Energy up to >20 mJ
- Small footprint
- Single phase 110 V or 230 V power, and no need for external cooling water
- All solid-state technology delivers rapid warm-up
- Fully computer-controlled

Spectra-Physics
A Division of Newport Corporation

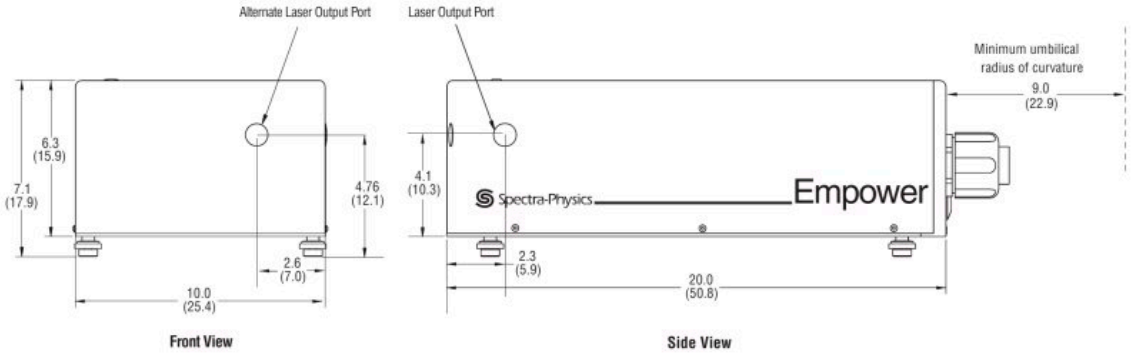
Specifications

General Characteristics[1]		Empower 15	Empower 30
Energy per Pulse	1 kHz	10.0 mJ	20.0 mJ
	5 kHz	3.0 mJ	6.0 mJ
Beam Diameter		3 mm (nominal)	
Energy Stability[2]		<1% rms	
Polarization		Linear, horizontal	
Transverse Mode[3]		Multimode	
Wavelength		527 nm	
Repetition Rate		1–10 kHz	
Utility Requirements			
Water		Closed-loop chiller included	
Electric		110/230 VAC, single phase, 60/50 Hz, 2 x 15/10 A	
Physical Characteristics			
Size (L x W x H)			
Laser Head		20 x 10 x 7.1 in (50.8 x 25.4 x 17.9 cm)	
Power Supply		18 x 19 x 7 in (45.47 x 48.26 x 17.53 cm)	
Weight			
Laser Head		50 lb (23 kg)	
Power Supply		45 lb (21 kg)	
Umbilical Length		10 ft (3 m)	

Notes

1. Due to our continuous product improvement program, specifications are subject to change without notice
2. rms with near Gaussian distribution
3. Optimized for pumping Ti:Sapphire lasers

Empower Laser Head



Dimensions in In (cm)

VISIBLE AND/OR INVISIBLE* LASER RADIATION
AVOID EYE OR SKIN EXPOSURE TO DIRECT OR SCATTERED RADIATION
CLASS 4 LASER PRODUCT
MAXIMUM OUTPUT IS DEPENDANT UPON AND LESS THAN INPUT RADIATION. EMITTED WAVELENGTH DEPENDS UPON FILTER SETTING.

RWTÜV USA INC. DAR TGA-ZQ-37-95-00-0
ISO 9001 Certified

Newport Experience | Solutions

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